DSA0405 -FUNDAMENTALS OF DATA SCIENCE

CO3 AT-1 CASE STUDY

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CO-3 AT_1

Case Study 6 – Detecting an Unusual Customer

Given Data

Particular	Value
Mean Spending (μ)	₹2,000
Standard Deviation (σ)	₹400
Customer Spending (x)	₹3,200

Question 1: Calculate the customer's z-score.

Formula

z=x−μσz=\frac{x-\mu}{\sigma}z=σx−μ

Where:

- **x** = Customer Spending
- **μ** = Mean Spending
- **σ** = Standard Deviation

Step-by-Step Calculation

Step 1: Substitute the given values into the formula.

z=3200−2000400z=\frac{3200-2000}{400}z=4003200−2000

Step 2: Subtract the numerator.

z=1200400z=\frac{1200}{400}z=4001200

Step 3: Divide the numerator by the denominator.

z=3z=3z=3

Answer

z=3\boxed{z=3}z=3

The customer's **z-score is 3**

Question 2: Is the customer's spending unusually high compared with the average?

Explanation

A **z-score** indicates how far a value is from the mean in terms of standard deviations.

General Interpretation of Z-score

Z-score	Interpretation
0	Equal to the average
±1	Close to the average
±2	Unusually high or low
±3 or above	Extremely unusual

Step-by-Step Analysis

From **Question 1**,

z=3z=3z=3

This means the customer spends

- **3 standard deviations above the average spending.**

Since a z-score greater than **2** is generally considered unusual,

3>23>23>2

Therefore,

the customer's spending is considered **unusually high**.

Answer

Yes.

The customer spends **₹3,200**, which is **3 standard deviations above the average spending of ₹2,000**. Therefore, the spending is **unusually high** compared with other customers.

Question 3: What would a negative z-score mean in this context?

Explanation

A **negative z-score** means the observed value is **less than the mean**.

In this case,

the observed value is the customer's spending.

If

z<0z<0z<0

then

x<μx<\mux<μ

This means

Customer Spending < Average Spending

Example

Suppose a customer spends **₹1,400**.

Then

z=1400−2000400z=\frac{1400-2000}{400}z=4001400−2000

Subtract the numerator.

z=−600400z=\frac{-600}{400}z=400−600

Divide.

z=−1.5z=-1.5z=−1.5

The z-score is negative because the customer spent **less than the average**.

Answer

A negative z-score indicates that the customer's spending is **below the average spending**. The farther the value is below zero, the lower the customer's spending compared with the average.

Question 4: If another customer has a z-score of −1.5, what does it tell you?

Given

z=−1.5z=-1.5z=−1.5

Step-by-Step Interpretation

A z-score of

−1.5-1.5−1.5

means

the customer spends

1.5 standard deviations below the average.

Since

−1.5<0-1.5<0−1.5<0

the spending is below the average.

However,

because the z-score is **greater than −2**, it is **not considered extremely unusual**.

Mathematical Interpretation

Average Spending

=₹2000=₹2000=₹2000

Difference from mean

1.5×400=₹6001.5\times400=₹6001.5×400=₹600

Customer Spending

2000−600=₹14002000-600=₹14002000−600=₹1400

Therefore,

a customer with a z-score of **−1.5** spends approximately **₹1,400**.

Answer

A z-score of **−1.5** means the customer spends approximately **₹1,400**, which is **1.5 standard deviations below the average**. The customer spends less than the average but is **not considered an extreme outlier**.

Summary Table

Question	Result
Mean Spending	₹2,000
Standard Deviation	₹400
Customer Spending	₹3,200
Z-score	**3**
Spending Level	Unusually High
Negative Z-score	Spending below average
Z-score = −1.5	About ₹1,400 (1.5 standard deviations below the mean)

Case Study 7 – Delivery Time Analysis

Given Data

The delivery times (in minutes) are:

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Question 1: Calculate the Mean and Median.

Formula for Mean

Mean=∑XN\text{Mean}=\frac{\sum X}{N}Mean=N∑X

Where:

- ∑X\sum X∑X = Sum of all observations
- NNN = Total number of observations

Step-by-Step Calculation of Mean

Step 1: Find the sum of all delivery times.

Delivery Time (Minutes)
25
27
28
29
30
30
31
32
34
60

∑X=25+27+28+29+30+30+31+32+34+60\sum
X=25+27+28+29+30+30+31+32+34+60∑X=25+27+28+29+30+30+31+32+34+60
∑X=326\sum X=326∑X=326

Step 2: Count the total number of observations.

N=10N=10N=10

Step 3: Apply the formula.

Mean=32610\text{Mean}=\frac{326}{10}Mean=10326
Mean=32.6\text{Mean}=32.6Mean=32.6

Answer (Mean)

Mean=32.6 minutes\boxed{\text{Mean}=32.6\text{ minutes}}Mean=32.6 minutes

Formula for Median

For an **even** number of observations,

Median=(N2)th value+(N2+1)th value2\text{Median}=\frac{\left(\frac{N}{2}\right)^{th}\text{ value}+\left(\frac{N}{2}+1\right)^{th}\text{ value}}{2}Median=2(2N)th value+(2N +1)th value

Step-by-Step Calculation of Median

Step 1: Arrange the data in ascending order.

Position	Value
1	25
2	27
3	28
4	29
5	30
6	30
7	31
8	32
9	34
10	60

Step 2: Count the observations.

N=10N=10N=10

Since **N is even**, the median is the average of the **5th and 6th observations**.

5th observation = **30**

6th observation = **30**

Step 3: Apply the formula.

$$\text{Median}=\frac{30+30}{2}$$
$$\text{Median}=30$$

Answer (Median)

$$\boxed{\text{Median}=30\text{ minutes}}$$

Question 2: Find Q1 and Q3.

Formula

- **Q1** = Median of the lower half of the data.
- **Q3** = Median of the upper half of the data.

Step-by-Step Calculation

Step 1: Divide the data into two halves.

Lower Half

25, 27, 28, 29, 30

Upper Half

30, 31, 32, 34, 60

Step 2: Find Q1.

The lower half contains **5 observations**.

Median position

$$\frac{5+1}{2}=3^{rd}$$

Third value

= **28**

Therefore,

$$Q_1=28$$

Step 3: Find Q3.

Upper half

30,31,32,34,60

Median position

5+12=3rd\frac{5+1}{2}=3^{rd}25+1=3rd

Third value

= **32**

Therefore,

Q3=32Q_3=32Q3=32

Answer

Q1=28\boxed{Q_1=28}Q1=28 Q3=32\boxed{Q_3=32}Q3=32

Question 3: Calculate the Interquartile Range (IQR).

Formula

IQR=Q3−Q1IQR=Q_3-Q_1IQR=Q3−Q1

Step-by-Step Calculation

Substitute the values.

IQR=32−28IQR=32-28IQR=32−28 IQR=4IQR=4IQR=4

Answer

IQR=4\boxed{IQR=4}IQR=4

Question 4: Check for Outliers Using the 1.5 × IQR Rule.

Formula

Lower Limit

Q1−1.5(IQR)Q_1-1.5(IQR)Q1−1.5(IQR)

Upper Limit

Q3+1.5(IQR)Q_3+1.5(IQR)Q3+1.5(IQR)

Step 1: Calculate 1.5 × IQR

1.5×4=61.5\times4=61.5×4=6

Step 2: Calculate the Lower Limit.

28−6=2228-6=2228−6=22

Lower Limit

=22=22=22

Step 3: Calculate the Upper Limit.

32+6=3832+6=3832+6=38

Upper Limit

=38=38=38

Step 4: Compare every value with the limits.

Value	Within Limit?
25	Yes
27	Yes
28	Yes
29	Yes
30	Yes
30	Yes
31	Yes
32	Yes
34	Yes
60	**No**

Since

60>3860>3860>38

60 is an **outlier**.

Answer

The value **60 minutes** is an **outlier** because it is greater than the upper limit of **38 minutes**.

Question 5: Explain whether the Mean or Median gives a better description of the typical delivery time.

Explanation

The mean uses **every value** in the dataset.

Because **60 minutes** is much larger than the other delivery times, it increases the mean.

Mean

= **32.6 minutes**

Median

= **30 minutes**

The median is **not affected** by extreme values.

Since the dataset contains an outlier, the **median** represents the typical delivery time more accurately.

Answer

The **median** is a better measure of the typical delivery time because it is **not influenced by the outlier (60 minutes)**, whereas the mean is increased by the unusually long delivery time.

Summary Table

Measure	Value
Mean	32.6 minutes
Median	30 minutes
Q1	28
Q3	32
IQR	4
Lower Limit	22
Upper Limit	38
Outlier	60 minutes

Step 1: Calculate the Mean

Original amounts
Equal shares

2
4
7
x‾=2+4+73=133=4 13\bar{x} = \frac{2 + 4 + 7}{3} = \frac{13}{3} = 4\ \frac{1}{3}x‾=32+4+7=313=4 31
x̄ is the arithmetic mean (simple average)
x1x_1x1
2
x1x_1x1
x2x_2x2
4
x2x_2x2
x3x_3x3
7
x3x_3x3

Formula

Mean=∑XN\text{Mean}=\frac{\sum X}{N}Mean=N∑X

where:

- ∑X\sum X∑X = Sum of all values
- NNN = Number of observations

City A

Add all temperatures

28+29+30+30+31+30+32=21028+29+30+30+31+30+32=21028+29+30+30+31+30+32=210

Number of days

N=7N=7N=7

Mean

Mean=2107=30∘C\text{Mean}=\frac{210}{7}=30^\circ CMean=7210=30∘C

Mean of City A = 30°C

City B

Add all temperatures

20+25+30+35+30+25+45=21020+25+30+35+30+25+45=21020+25+30+35+30+25+45=210

Number of days

N=7N=7N=7

Mean

Mean=2107=30∘C\text{Mean}=\frac{210}{7}=30^\circ CMean=7210=30∘C

Mean of City B = 30°C

Step 2: Calculate the Variance

Var(X)=E(X2)−[E(X)]2Var(X) = E(X^2) - [E(X)]^2Var(X)=E(X2)−[E(X)]2
Var(X)=(1.4)2=1.96Var(X)=(\text{1.4})^2=\text{1.96}Var(X)=(1.4)2=1.96
σ\sigmaσ
1.
σ\sigmaσ
μ-σ+σVar(X) ≈ 1.96

Formula

Variance=∑(X−X‾)2N\text{Variance}=\frac{\sum (X-\bar X)^2}{N}Variance=N∑(X−X‾)2

where:

- XXX = Each value
- X‾\bar XX‾ = Mean
- NNN = Number of observations

City A

Mean = **30**

Temperature (X)	X − Mean	(X − Mean)²
28	28−30 = -2	4
29	29−30 = -1	1
30	30−30 = 0	0
30	30−30 = 0	0
31	31−30 = 1	1
30	30−30 = 0	0
32	32−30 = 2	4

Step 1: Add squared deviations

4+1+0+0+1+0+4=104+1+0+0+1+0+4=104+1+0+0+1+0+4=10

Step 2: Divide by number of observations

Variance=107=1.43\text{Variance}=\frac{10}{7}=1.43Variance=710=1.43

Variance of City A = 1.43

City B

Mean = **30**

Temperature (X)	**X − Mean**	**(X − Mean)^2**
20	20−30 = -10	100
25	25−30 = -5	25
30	30−30 = 0	0
35	35−30 = 5	25
30	30−30 = 0	0
25	25−30 = -5	25
45	45−30 = 15	225

Step 1: Add squared deviations

100+25+0+25+0+25+225=400100+25+0+25+0+25+225=400100+25+0+25+0+25+225=400

Step 2: Divide by number of observations

Variance=4007=57.14\text{Variance}=\frac{400}{7}=57.14Variance=7400=57.14

Variance of City B = 57.14

Step 3: Compare the Variance

- **City A Variance = 1.43**
- **City B Variance = 57.14**

Since:

57.14>1.4357.14 > 1.4357.14>1.43

Answer: City B has greater temperature variability.

Step 4: Why can two datasets have the same mean but different standard deviations?

- Both cities have the same **mean (30°C)**.

- However, **City A's temperatures stay close to 30°C**.
- **City B's temperatures are spread out between 20°C and 45°C**.
- Therefore, **the averages are the same, but the spread is different**, resulting in different standard deviations and variances.

Step 5: Which city has more predictable temperatures

A city with **lower variance** has more consistent temperatures.

- City A Variance = **1.43**
- City B Variance = **57.14**

Since **1.43 is much smaller than 57.14**, **City A has more predictable temperatures.**

Final Answers

1. **Mean Temperature**
   - City A = **30°C**
   - City B = **30°C**
2. **Variance**
   - City A = **1.43 °C²**
   - City B = **57.14 °C²**
3. **Greater temperature variability: City B**
4. **Reason:** The mean gives the average, while the standard deviation and variance measure how spread out the data are. Two datasets can have the same average but different spreads.
5. **More predictable temperatures: City A**

Data Center CPU Usage A data scientist records CPU usage percentages: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90 1. Calculate mean, median, and mode. 2. Calculate the range. 3. Find Q1, Q3 and IQR. 4. Identify possible outliers. 5. Determine the direction of skewness.give step by step process

Data Center CPU Usage

Given Data (Already in Ascending Order)

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Number of observations:

n=10n=10n=10

1. Calculate Mean, Median, and Mode

Step 1: Mean

Original amounts

Equal shares
2
4
7
x¯=2+4+73=133=4 13\bar{x} = \frac{2 + 4 + 7}{3} = \frac{13}{3} = 4\
\frac{1}{3}x¯=32+4+7=313=4 31
x̄ is the arithmetic mean (simple average)
x1x_1x1
2
x1x_1x1
x2x_2x2
4
x2x_2x2
x3x_3x3
7
x3x_3x3

Formula

Mean=∑Xn\text{Mean}=\frac{\sum X}{n}Mean=n∑X

Add all values

40+42+45+45+46+48+50+52+55+90=51340+42+45+45+46+48+50+52+55+90=51340+42+45+45+46+48+50+52+55+90=513

Divide by number of observations

Mean=51310=51.3\text{Mean}=\frac{513}{10}=51.3Mean=10513=51.3

Mean = 51.3

Step 2: Median

Formula

For even number of observations,

Median=(n2)th+(n2+1)th2\text{Median}=\frac{\left(\frac{n}{2}\right)^{th}+\left(\frac{n}{2}+1\right)^{th}}{2}Median=2(2n)th+(2n+1)th

Since

n=10n=10n=10

The middle numbers are

- 5th value = 46
- 6th value = 48

$$\text{Median}=\frac{46+48}{2}=\frac{94}{2}=47$$

Median = 47

Step 3: Mode

Mode is the value occurring most frequently.

Data:

40, 42, **45, 45**, 46, 48, 50, 52, 55, 90

45 appears **2 times**.

Mode = 45

2. Calculate the Range

Formula

$$\text{Range}=\text{Maximum}-\text{Minimum}$$

Maximum = 90

Minimum = 40

$$\text{Range}=90-40=50$$

Range = 50

3. Find Q1, Q3 and IQR

Step 1: Divide the data into two halves

Lower Half

40, 42, 45, 45, 46

Upper Half

48, 50, 52, 55, 90

Step 2: Find Q1

Middle value of lower half

40, 42, **45**, 45, 46

Q1=45Q_1=45Q1=45

Step 3: Find Q3

Middle value of upper half

48, 50, **52**, 55, 90

Q3=52Q_3=52Q3=52

Step 4: Calculate IQR

Formula

IQR=Q3−Q1IQR=Q_3-Q_1IQR=Q3−Q1 IQR=52−45=7IQR=52-45=7IQR=52−45=7

IQR = 7

4. Identify Possible Outliers

Step 1: Calculate Lower Fence

Lower Fence=Q1−1.5(IQR)\text{Lower Fence}=Q_1-1.5(IQR)Lower Fence=Q1−1.5(IQR) =45−1.5(7)=45-1.5(7)=45−1.5(7) =45−10.5=45-10.5=45−10.5 =34.5=34.5=34.5

Step 2: Calculate Upper Fence

Upper Fence=Q3+1.5(IQR)\text{Upper Fence}=Q_3+1.5(IQR)Upper Fence=Q3+1.5(IQR) =52+1.5(7)=52+1.5(7)=52+1.5(7) =52+10.5=52+10.5=52+10.5 =62.5=62.5=62.5

Step 3: Check the data

Lower Fence = **34.5**

Upper Fence = **62.5**

Data:

40, 42, 45, 45, 46, 48, 50, 52, 55, **90**

Since

90>62.590>62.590>62.5

90 is an outlier.

5. Determine the Direction of Skewness

-6-4-22460.10.20.30.4ValueDensityMeanMedianMode
Right skew: mode, median, mean from peak to tail
Direction
LeftRight
LeftRight
Skew strength
0.
Skew strength

Compare the mean and median:

- Mean = **51.3**
- Median = **47**

Since

Mean>Median\text{Mean}>\text{Median}Mean>Median

the distribution is **positively skewed (right-skewed)**.

This is because the high value **90** pulls the mean toward the right.

Final Answers

Question	Answer
Mean	**51.3**
Median	**47**
Mode	**45**
Range	**50**
Q1	**45**
Q3	**52**
IQR	**7**
Outlier	**90**
Skewness	**Positively (Right) Skewed**

Comparing Two Datasets Two data analysts receive the following datasets: Dataset A: 10, 20, 30, 40, 50 Dataset B: 28, 29, 30, 31, 32 1. Calculate the mean of both datasets. 2. Calculate the variance of both. 3. Calculate the standard deviation of both. 4. Which dataset has greater spread? 5. If both datasets represent model errors, which model appears more consistent? 6. Explain why mean alone is not enough to compare datasets.give step by step process

Comparing Two Datasets – Step-by-Step Solution

Given Data

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

Number of observations:

n=5n=5n=5

1. Calculate the Mean

Original amounts
Equal shares
2
4
7
x‾=2+4+73=133=4 13\bar{x} = \frac{2 + 4 + 7}{3} = \frac{13}{3} = 4\
\frac{1}{3}x‾=32+4+7=313=4 31
x̄ is the arithmetic mean (simple average)
x1x_1x1
2
x1x_1x1
x2x_2x2
4
x2x_2x2
x3x_3x3
7
x3x_3x3

Formula

Mean=∑Xn\text{Mean}=\frac{\sum X}{n}Mean=n∑X

Dataset

Step 1: Add all values

10+20+30+40+50=15010+20+30+40+50=15010+20+30+40+50=150

Step 2: Divide by number of observations

Mean=1505=30\text{Mean}=\frac{150}{5}=30Mean=5150=30

Mean of Dataset A = 30

Dataset B

Step 1: Add all values

28+29+30+31+32=15028+29+30+31+32=15028+29+30+31+32=150

Step 2: Divide by number of observations

Mean=1505=30\text{Mean}=\frac{150}{5}=30Mean=5150=30

Mean of Dataset B = 30

2. Calculate the Varianc

Var(X)=E(X2)−[E(X)]2Var(X) = E(X^2) - [E(X)]^2Var(X)=E(X2)−[E(X)]2
Var(X)=(1.4)2=1.96Var(X)=(\text{1.4})^2=\text{1.96}Var(X)=(1.4)2=1.96
σ\sigmaσ
1.
σ\sigmaσ
μ-σ+σVar(X) ≈ 1.96

Formula

Variance=∑(X−X‾)2n\text{Variance}=\frac{\sum (X-\bar X)^2}{n}Variance=n∑(X−X‾)2

where

- XXX = Each value
- X‾\bar XX‾ = Mean
- nnn = Number of observations

Dataset A

Mean = **30**

X	**X − Mean**	**$(X - \text{Mean})^2$**
10	−20	400
20	−10	100
30	0	0
40	10	100
50	20	400

Step 1: Add squared deviations

400+100+0+100+400=1000400+100+0+100+400=1000400+100+0+100+400=1000

Step 2: Divide by 5

Variance=10005=200\text{Variance}=\frac{1000}{5}=200Variance=51000=200

Variance of Dataset A = 200

Dataset B

Mean = **30**

X	X − Mean	(X − Mean)²
28	−2	4
29	−1	1
30	0	0
31	1	1
32	2	4

Step 1: Add squared deviations

4+1+0+1+4=104+1+0+1+4=104+1+0+1+4=10

Step 2: Divide by 5

Variance=105=2\text{Variance}=\frac{10}{5}=2Variance=510=2

Variance of Dataset B = 2

3. Calculate the Standard Deviation

σ=1N∑i=1N(xi−μ)2\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2}σ=N1∑i=1N(xi−μ)2
The points have mean μ = 5 and σ = 1.5.
σ\sigmaσ
1.5
σ\sigmaσ
Drag points

Formula

Standard Deviation=Variance\text{Standard Deviation}=\sqrt{\text{Variance}}Standard Deviation=Variance

Dataset A

Variance = **200**

SD=200SD=\sqrt{200}SD=200 SD≈14.14SD\approx14.14SD≈14.14

Standard Deviation = 14.14

Dataset B

Variance = **2**

SD=2SD=\sqrt2SD=2 SD≈1.41SD\approx1.41SD≈1.41

Standard Deviation = 1.41

4. Which Dataset Has Greater Spread?

Compare the standard deviations.

Dataset A

SD=14.14SD=14.14SD=14.14

Dataset B

SD=1.41SD=1.41SD=1.41

Since

14.14>1.4114.14>1.4114.14>1.41

Dataset A has greater spread (higher variability).

5. If Both Datasets Represent Model Errors, Which Model Appears More Consistent?

A model with a **smaller standard deviation** has errors that are closer to the mean.

- Dataset A: SD = **14.14**
- Dataset B: SD = **1.41**

Since **1.41 is much smaller**, **Dataset B represents the more consistent model.**

6. Why Is Mean Alone Not Enough to Compare Datasets?

Both datasets have the **same mean (30)**.

However:

- Dataset A values are spread widely from **10 to 50**.
- Dataset B values are close together between **28 and 32**.

The **mean only tells the average**, not how spread out the values are. **Variance and standard deviation** measure the spread, making them essential for comparing datasets with the same mean.